Antibody assessment for the lipoplex — from the wet-lab suggestion

Source: `LIPOPLEX(1).docx`. The document's antibody content is a **targeting** proposal (surface ligands for neurotropism), not validation reagents. Assessed below against the Antibody Registry, UniProt topology, and the project's own engineered insert.

THE BLOCKING FINDING: no commercial antibody can detect this construct's product

`engPENK_insert_annotated.gb` (885 nt, CDS 13..879, 288 aa ORF) encodes **7 identical copies of the octapeptide `YGGFEVVG`**, each flanked by dibasic KR/KK processing sites.

Native human PENK (P01210, 267 aa) carries 6x `YGGFM` + 1x `YGGFL` [2,3]. The engineered construct carries **0 copies of either**.

Every anti-enkephalin antibody in the registry is raised against the native mature pentapeptide, so all of them miss the product:

RRID	antibody	epitope	verdict
AB_2530349	anti-Met-enkephalin mAb 1194/334	YGGFM	**will not detect**
AB_2492947	anti-Met-enkephalin pAb	PENK 237–241 (=YGGFM)	**will not detect**
AB_3430419	anti-Met-enkephalin	Met-enkephalin	**will not detect**
—	custom anti-`YGGFEVVG`–KLH	the engineered octapeptide	**required; must be custom-raised**
—	precursor-directed anti-PENK	PENK backbone	precursor only; cannot show maturation

The shared `YGGF` N-terminal core is not sufficient: anti-Met-enkephalin antibodies are raised against the whole pentapeptide and typically require the free C-terminal Met, which the product does not have. **Order a custom polyclonal against `YGGFEVVG` (KLH-conjugated, ≥2 rabbits) now** — it is a 10–12 week lead time and it gates the entire expression readout.

Targeting: TfR is sound, TRPV1 is weak

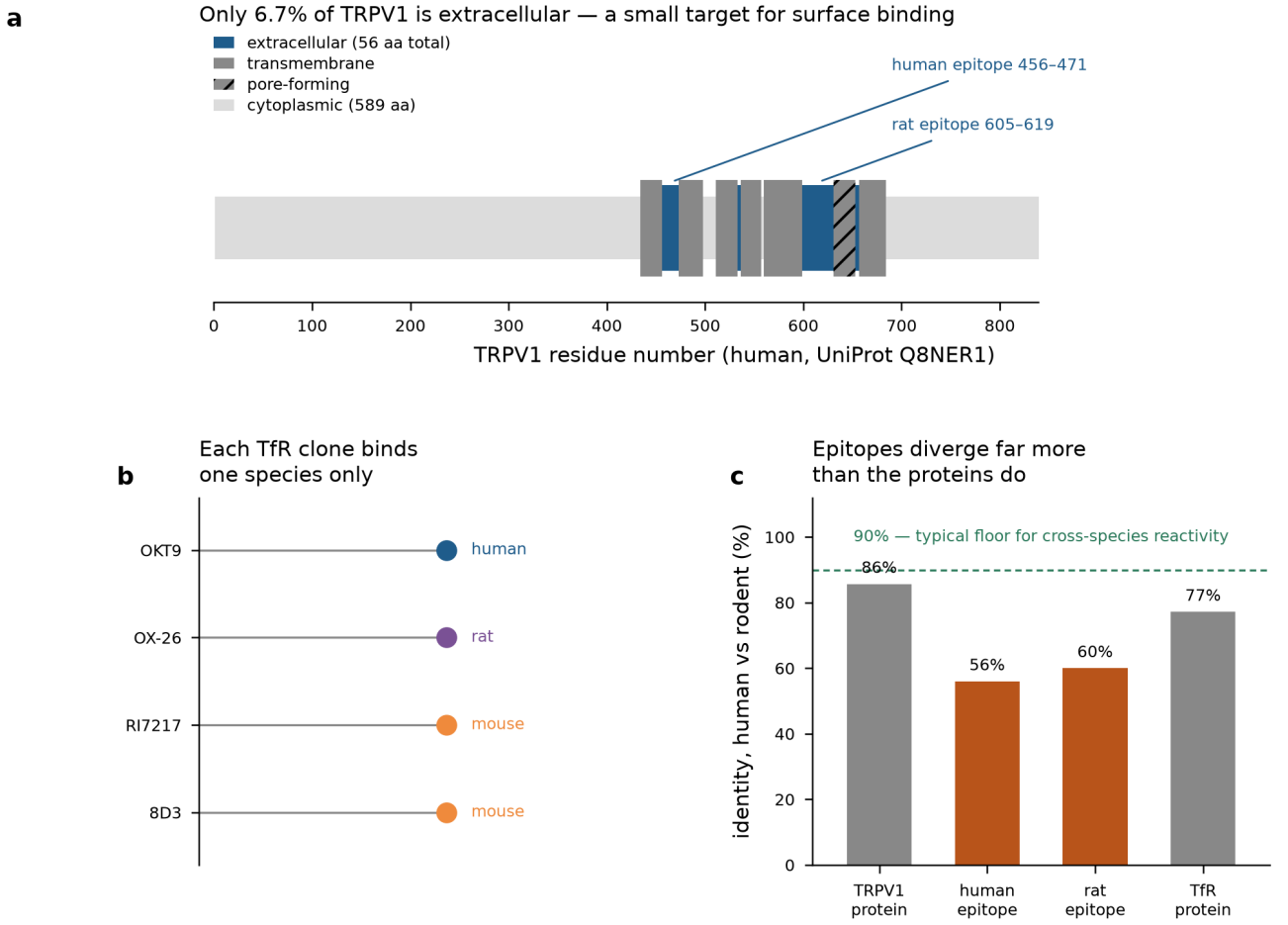


Figure 1. Epitope accessibility and species restriction of the proposed targeting antibodies. (a) Membrane topology of human TRPV1 (UniProt Q8NER1, 839 aa) from curated topological-domain annotation. Only four extracellular loops totalling 56 aa (6.7% of the protein) are reachable by an antibody on an intact cell; both commercially “extracellular” epitopes fall inside them, but the loops are only 17 and 32 aa. (b) Each transferrin-receptor clone in the Antibody Registry binds one species; n = 1 clone per row. (c) Percent identity between human and rodent orthologue by global Needleman–Wunsch alignment (match +2, mismatch −1, gap −1). Epitope bars are the 16 aa and 15 aa vendor peptides against their best-matching window in the other species. Whole proteins are 77–86% identical but the epitopes only 56–60%, which is why neither TRPV1 reagent transfers between models; cross-species reactivity is generally retained only above ~90% (dashed line).

TfR (the document's own preferred target). The transferrin receptor has been the canonical brain-endothelium target since it was localised there [4], and antibody–drug conjugates against it cross the barrier [5]. Verified in the registry; each clone is strictly species-restricted, so the reagent must be chosen for the model, not the target:

clone	RRID	binds	note
OKT9	AB_3714951	human TFRC (P02786)	human cells only
OX-26	AB_2573153	rat TFRC (Q99376)	the classic BBB-transcytosis clone
RI7217	AB_3715099	mouse TFRC (Q62351)	most-cited low-avidity transcytosis clone
8D3	AB_1511006	mouse TFRC	high avidity; binds BECs but transcytoses poorly

Human vs mouse TfR is 77.2% identical (global alignment) — below the ~90% where cross-reactivity is usually retained. There is **no single TfR antibody that works in both the rodent model and human**

cells. The document's point about preferring low affinity/avidity for transcytosis is correct [7,12] and is the reason to prefer RI7217 over 8D3.

TRPV1 (`anti-TRPV1` in the document) is a poor surface-targeting choice. Two registry antibodies are sold as "extracellular" and both check out against UniProt topology [16], consistent with the cryo-EM structures [9,10] — human aa 456–471 `RPVDGLPPFKMEKTGD` sits in the 456–472 extracellular domain, and rat aa 605–619 in the 599–630 loop. But:

- **Only 56 of 839 residues (6.7%) of TRPV1 are extracellular**, in four loops of 17, 3, 32 and 4 aa. There is very little surface to bind.
- Both are **polyclonal** — no orientation control for site-directed conjugation.
- The epitopes are only 56–60% conserved human↔rat, against 85.7% for the whole protein, so neither reagent transfers between models.
- The rat vendor peptide differs from UniProt O35433 at one position (Cys616→Ser); confirm with the vendor, as a Cys in the immunogen is also a conjugation handle.

Only 15 of 485 TRPV1 records mention flow cytometry; the set is dominated by western (194) and IHC (145) — i.e. fixed/permeabilised formats that say nothing about live-cell surface accessibility.

Recommendation: keep the RVG peptide as the neuronal ligand [6] (it is a peptide, cheap, and needs no antibody), use TfR for BBB transcytosis, and drop anti-TRPV1 as a targeting ligand. If TRPV1 selectivity is wanted, a small-molecule TRPV1 ligand is the better handle than an antibody against a 6.7%-exposed target.

Formulation QC (available now, cheap)

clone	RRID	detects
RM105	AB_2716366	mPEG methoxy terminus
B141M / B139M	AB_936779 / AB_936777	PEG backbone

Terminus + backbone pair supports a sandwich ELISA for surface PEG density, and screens anti-PEG immunogenicity [14] — a real failure mode for the repeat-dosed PEG(2000)-CerC8 formulation the document specifies.

siRNA arm — re-designed for human exclusivity

The wet lab requires the siRNA to be **exclusively human, with no mouse cross-reactivity.** The previously delivered panel does not meet that constraint. Every one of its 11 guides is a perfect 19/19 match to human MME (NM_000902.5), but tested against mouse Mme (NM_008604.4) **none is exclusive**: 8 retain ≥17/19 identity and two (CDS1590, CDS1616) are perfect 19/19 matches in mouse, i.e. fully cross-reactive.

Re-screening the whole human CDS (751 codons, 2253 nt) under the new constraint gives 307 composition- and asymmetry-filtered 19-mers, of which **26 fall to ≤15/19 against mouse.** Ranking those by seed divergence, human off-target burden (BLASTn vs human refseq_rna, word size 7, RID 6N566XEC014), duplex asymmetry [8] and GC content:

rank	guide (5'→3')	CDS pos	GC%	mouse	seed MM	off-targets ≥17/19	composite
1	AAAGUUGCCGUAACGGGAGdTdT	303	52.6	14/19	2	0	85.8
2	UCCUCGACCACAUGUUUACdTdT	1307	47.4	14/19	2	4	74.3
3	UGGAGUUAACCGCAUACUCdTdT	2101	47.4	15/19	3	0	61.6
4	UACAAGACCUGUACAAUGCdTdT	412	42.1	15/19	2	1	57.8
5	UGAAUUACAUGAUUCACAGdTdT	635	31.6	14/19	3	84	54.5

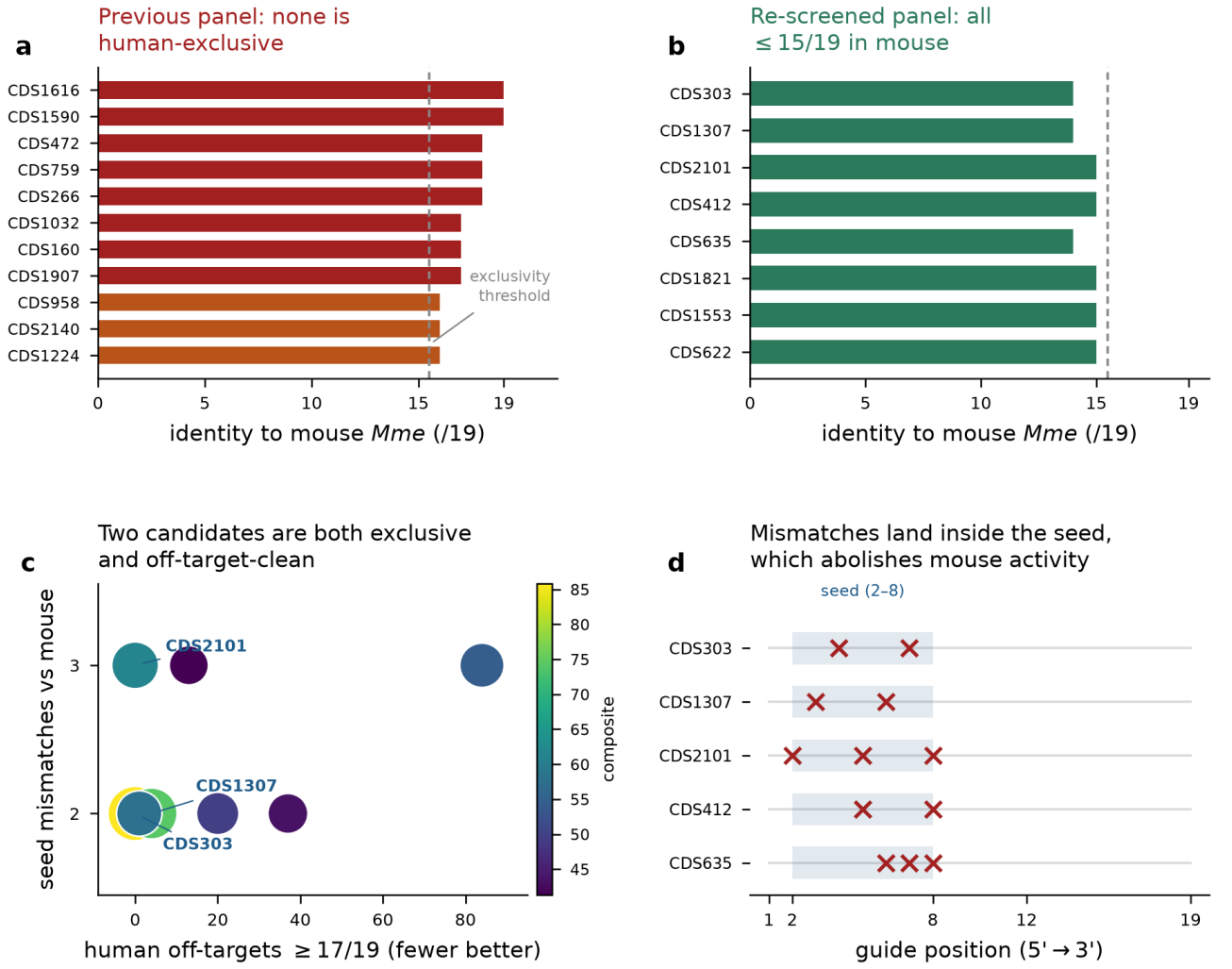


Figure 2. Enforcing human-exclusive targeting on the anti-MME siRNA panel. (a) The 11 previously delivered guides scored against mouse Mme (NM_008604.4); all are 19/19 on human MME, but none reaches the ≤15/19 exclusivity threshold (dashed) and CDS1590 and CDS1616 are perfect matches in mouse. Red ≥17/19, orange 16/19. (b) The eight top-ranked guides from re-screening the whole human CDS, all ≤15/19 in mouse. (c) Human off-target burden (BLASTn, refseq_rna, word size 7, gapless HSPs, MME excluded) against mouse seed divergence; marker area and colour both encode the composite score. CDS303 and CDS2101 combine exclusivity with zero off-targets at ≥17/19. (d) Positions of guide–mouse mismatches for the top five candidates; shading marks the seed (guide positions 2–8), where a mismatch most reliably abolishes activity on an unintended transcript.

Recommended lead: CDS303 — guide `AAAGUUGCCGUAACGGGAGdTdT`, passenger `CUCCCGUUACGGCAACUUUdTdT`. 19/19 on human MME, 14/19 on mouse Mme with 2 of those mismatches inside the seed (guide positions 4,7), **zero** human off-targets at ≥17/19, and favourable duplex asymmetry (ΔΔG = 4.20 kcal/mol, guide 5'-end less stable, which loads the correct strand). CDS2101 is

the best three-seed-mismatch alternative and also has zero ≥17/19 off-targets; order both plus a scrambled control.

Guide positions 2–8 are the seed; a mismatch there is the most reliable way to abolish activity on an unintended transcript, which is why exclusivity was ranked on seed divergence rather than on total identity alone. Note the corollary: **these guides cannot be tested in a mouse model.** Knockdown validation must be done in human cells, and any in-vivo rodent work needs a separate mouse-targeting surrogate — accepting that it is then a different molecule, not the same one.

For the validation western, only **4 MME/neprilysin records** exist in the registry; AB_2860234 (mouse mAb, CD10) is the lead. Confirm human reactivity and WB validation before ordering.

Note on the document's sources

Two links are NotebookLM notebooks (not citable) and one is a sci-hub mirror. That last one resolves to a legitimate citation: Shimoni, Shi, Adlard & Bush, "Delivery of Fluorescent Nanoparticles to the Brain", Journal of Molecular Neuroscience (2016), doi:10.1007/s12031-016-0833-5 (ref. 15) — cite that directly rather than the mirror.

Consequence for the CG-MD work

Adding a conjugated antibody is **not** a parameter change to the existing lipoplex model. An IgG is ~150 kDa against lipid components of a few hundred daltons, so it would dominate box size and runtime, and there is no coarse-grained membrane skill available here. If targeting ligands are going into the model, scFv or nanobody (as the document itself suggests, ~28/15 kDa) is the only tractable option — and the DOTAP/MC3 replicate results should be finished and reported on the bare formulation first.

What the wet lab still needs to supply

For each antibody they recommend: **catalog number + vendor, clone ID or RRID, the immunogen sequence or residue range, validated applications, species reactivity, and whether it is carrier-free.** The immunogen sequence is the one most often omitted and the one that decides whether a reagent works here — as the PENK finding above shows.

Methods

Reagent records were retrieved from the Antibody Registry and are cited by RRID [13]. Sequences and topological-domain annotation are from UniProt [16]: human TRPV1 Q8NER1, rat TRPV1 O35433, human TFRC P02786, mouse TFRC Q62351, rat TFRC Q99376, human PENK P01210. Transcripts are RefSeq NM_000902.5 (human MME) and NM_008604.4 (mouse Mme). Percent identities are global Needleman–Wunsch alignments [1] (match +2, mismatch −1, gap −1); the engineered insert was translated from `engPENK_insert_annotated.gb` (CDS 13..879). siRNA off-target screening used NCBI BLASTn against human refseq_rna (word size 7, no low-complexity filter, Homo sapiens restricted); duplex asymmetry is the nearest-neighbour ΔG37 difference between the guide 5'- and 3'-terminal tetramers. Every DOI below was verified against Crossref rather than recalled. Lipid- and targeted-nanoparticle formulations for formulation context: refs 8, 11.

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